STACKS, MODULI OF CURVES: LEMMA-IN-SMOOTH-LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1626-1634 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 1636-1678, the complete author proof (41 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_moduli-curves.json (pb=2035, stmt 1626-1634, proof 1636-1678).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DSS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 655–658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06AG: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1678–1678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ADD: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1678–1678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1678–1678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1678–1678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above the following are equivalent*

(1) *the classifying morphism* $\mathrm{Spec}(k) \to \mathcal{C}urves$ *factors through the open where* $\mathcal{C}urves \to \mathrm{Spec}(\mathbf{Z})$ *is smooth,*

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(2) *the deformation category $\mathcal{D}ef_X$ is unobstructed.*

Proof. Since $\mathcal{C}urves \longrightarrow \mathrm{Spec}(\mathbf{Z})$ is locally of finite presentation (Lemma 0DSS) formation of the open substack where $\mathcal{C}urves \longrightarrow \mathrm{Spec}(\mathbf{Z})$ is smooth commutes with flat base change (Morphisms of Stacks, Lemma 0DZR). Since the Cohen ring Λ is flat over $\mathbf{Z}$, we may work over Λ. In other words, we are trying to prove that

$$\Lambda\text{-}\mathcal{C}urves \longrightarrow \mathrm{Spec}(\Lambda)$$

is smooth in an open neighbourhood of the point $x_0 : \mathrm{Spec}(k) \to \Lambda\text{-}\mathcal{C}urves$ defined by X/k if and only if $\mathcal{D}ef_X$ is unobstructed.

The lemma now follows from Geometry of Stacks, Lemma 0DZS and the equality

$$\mathcal{D}ef_X = \mathcal{F}_{\Lambda\text{-}\mathcal{C}urves,k,x_0}$$

This equality is not completely trivial to establish. Namely, on the left hand side we have the deformation category classifying all flat deformations $Y \to \mathrm{Spec}(A)$ of X as a scheme over $A \in \mathrm{Ob}(\mathcal{C}_\Lambda)$. On the right hand side we have the deformation category classifying all flat morphisms $Y \to \mathrm{Spec}(A)$ with special fibre X where Y is an algebraic space and $Y \to \mathrm{Spec}(A)$ is proper, of finite presentation, and of relative dimension ≤ 1. Since A is Artinian, we find that Y is a scheme for example by Spaces over Fields, Lemma 0ADD. Thus it remains to show: a flat deformation $Y \to \mathrm{Spec}(A)$ of X as a scheme over an Artinian local ring A with residue field k is proper, of finite presentation, and of relative dimension ≤ 1. Relative dimension is defined in terms of fibres and hence holds automatically for Y/A since it holds for X/k. The morphism $Y \to \mathrm{Spec}(A)$ is proper and locally of finite presentation as this is true for $X \to \mathrm{Spec}(k)$, see More on Morphisms, Lemma 06AG. □